MILITARY SPECIFICATION

POWER SUPPLY, TELETYPEWRITER
PP4473( )/UG

1. SCOPE

1.1 This specification covers requirements for a power supply unit capable of delivering 12 amperes of regulated direct current (dc) for high level (130 volt (v)) teletypewriter signal circuits.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 — Box, Fiberboard
PPP-B-640 — Box, Fiberboard, Corrugated, Triple-Wall
PPP-B-001163 — Box, Corrugated Fiberboard, High Compression Strength, Weather-Resistant, Wax and Resin Impregnated
PPP-T-76 — Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (for carton sealing)
PPP-T-97 — Tape, Pressure-Sensitive Adhesive, Filament Reinforced

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MIL-P-116 — Preservation, Methods of
MIL-D-3464 — Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-Q-9858 — Quality Program Requirements
MIL-M-15071 — Manuals, Equipment and Systems
MIL-E-16400 — Electronic Equipment, Naval Ship and Shore, General Specification
MIL-E-17555 — Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-F-22191 — Films, Transparent, Flexible, Heat Sealable, (for packaging applications)
MIL-D-23140 — Drawings, Preliminary and Final, Electronic Equipment and Systems, Installation Control
MIL-B-43291 — Box, Fireboard, Corrugated, Double-Wall, Weather Resistant

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108 — Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipments
MIL-STD-189 — Racks, Electrical Equipment, 19-inch and Associated Panels
MIL-STD-454 — Standard General Requirements for Electronic Equipment
MIL-STD-461 — Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-701 — List of Standard Semiconductor Devices
MIL-STD-781 — Reliability Tests: Exponential Distribution
MIL-STD-785 — Requirements for a Reliability Program (For Systems and Equipments)
MIL-STD-810 — Environmental Test Methods for Aerospace and Ground Equipment

PUBLICATIONS

MILITARY

MIL-HDBK-217 — Reliability Stress and Failure Rate Data for Electronic Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Ave. at 33rd Street, New York, N. Y. 10016).

3. REQUIREMENTS

3.1 First article sample.- Prior to beginning production a sample shall be inspected as specified in 4.3 (see 6.3).

3.2 Design objectives.- The equipment design shall be such as to attain the basic objectives of MIL-E-16400, as applicable to shore installations.

3.3 Reliability.-

3.3.1 Quantitative reliability requirement.- The specified mean-time-between-failure (MTBF) (θ_0 as defined by MIL-STD-781) of this equipment shall be 10,000 hours.

3.3.2 Reliability program plan.- The contractor shall submit a reliability program plan in accordance with and in paragraph sequence with MIL-STD-785. Particular attention should be paid to those sections of MIL-STD-785 which require submission of detailed tasks, procedures to implement and control these tasks, and time schedule. The proposed reliability program plan shall be submitted as a separate and complete entity within the total system or equipment project delineation. The reliability program plan as approved and incorporated into the contract becomes the basis for contractual compliance. The plan, including reliability test plans, shall describe an integrated effort within the quantitative reliability requirements during development and manufacture. The final reliability program plan shall be submitted for Government approval.

3.3.3 Reliability predicition.- The contractor shall submit a reliability prediction in accordance with MIL-HDBK-217 to the command or agency concerned. The prediction shall assume a 50^0 C ambient temperature and a shore environment. The contractor shall evaluate the prediction with respect to the reliability expected from a production unit. The processes and procedures required to assure production of reliable equipment shall be thoroughly documented.

3.3.4 Source of failure rates.- The primary source of failure rate data shall be MIL-HDBK-217. For items not included in MIL-HDBK-217, other sources of data may be used with the approval of the command or agency concerned. When such alternate sources of data are proposed, enough information shall be submitted to enable the command or agency concerned to evaluate the validity and applicability of the results derived therefrom.

3.3.5 Reliability reports.- Reports providing a complete accounting of progress on each element defined by the reliability program plan, results achieved, and status of actions to resolve problems and correct weak links shall be submitted monthly to the command or agency concerned.

3.4 Design requirements.-

3.4.1 General design.- The unit shall consist basically of a solid state rectifier with regulating and control circuits to achieve the performance specified under 3.5.

3.4.2 Material and parts.- The material and parts used in the construction of this equipment shall be selected in accordance with MIL-E-16400. Rectifier and regulator elements shall be solid state, selected in accordance with MIL-STD-701. The contractor may elect to use nonstandard parts as specified in

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MIL-E-16400. Items or processes which require approval of the command or agency concerned are listed in MIL-E-16400.

3.4.3 Environmental conditions.- The equipment shall meet the temperature conditions requirements (class 4 of table II) of MIL-E-16400. The equipment shall also meet the humidity requirements of MIL-E-16400.

3.5 Electrical design.-

3.5.1 Primary power will be single phase 120 volts alternating current (ac) ±10 percent at 50 or 60 Hertz (Hz) ±5 percent.

3.5.2 Rated maximum output shall be 12 amperes at 130 volts dc.

3.5.2.1 Efficiency.- The power conversion efficiency (dc watts output/ac watts input) shall be not less than 70 percent at full rated load.

3.5.2.2 Line regulation.- Less than 1 percent variation of output voltage for the full range of primary voltage.

3.5.2.3 Load regulation.- Less than 1 percent variation of output voltage from no load to full load, with constant primary voltage.

3.5.2.4 Ripple in the output shall not exceed 0.5 percent when working into a resistive load for any load between zero and rated maximum output.

3.5.2.5 Parallel operation.- The power supplies shall be capable of parallel operation, configured so that upon failure of any unit in a parallel group, the remaining units will automatically take up the load. For such operation the maximum output may be derated not to exceed 10 percent.

3.5.3 The unit shall incorporate a potentiometer to permit adjustment of the dc output voltage from 115 to 135 when the primary power is within the limits stated in 3.5.1.

3.5.4 There shall be circuit breakers with indicators for disconnecting each lead of the input and output circuits. The equipment shall incorporate automatic protection against damage from overload. There shall also be protection of the output circuit in case of a failure in the equipment which might draw power from the dc line.

3.5.5 A power input switch shall be provided which will completely disconnect the ac input from the equipment when in the "OFF" position and a power output switch which will completely disconnect the output line when in the "OFF" position.

3.5.6 The power supplies shall be provided with a satisfactory ammeter and voltmeter to indicate output current and voltage. The voltmeter shall be suppressed zero type, with scale reading 100 to 150 volts dc.

3.5.7 The circuit shall be designed to permit either the positive or negative output terminal to be operated at ground potential.

3.5.8 Output overvoltage of 10 percent or undervoltage of 10 percent, referred to 130/115 volts, shall cause a contact closure satisfactory for actuation of an alarm. The contacts shall be capable of handling 200 volts at 500 milliamperes, dc.

3.5.9 A chassis ground terminal shall be provided.

3.6 Mechanical design.-

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3.6.1 Enclosure.- The power supply shall be designed to mount in a standard 19-inch relay rack in accordance with MIL-STD-189. It shall have a satisfactory cover with convenient access for inspection and replacement of parts. Degree of enclosure shall be as specified for "Open protected" in MIL-STD-108.

3.6.2 Size and weight.- Depth shall not exceed 20 inches. Height shall be a standard multiple of 1 3/4 inches. Weight shall be the minimum necessary to achieve the specified performance, reliability, and maintainability.

3.6.3 Connection to the power units shall be made to screw-type barrier terminals on the rear of the chassis. The terminal boards shall have insulated protective covers or other measures to minimize the hazard of exposed connections.

3.6.4 Front panel.- All controls, indicators, circuit breakers, and switches shall be mounted on the front panel. Suitable handles shall be provided on the front panel to facilitate installation and maintenance of the equipment.

3.6.4.1 A power-on indicator light with a red jewel shall be mounted on the panel.

3.6.5 Painting shall be in accordance with MIL-E-16400.

3.6.6 Test points.- Significant voltages, currents, and wave forms shall be measurable at test points to facilitate alignment and determination of circuit faults.

3.7 Electromagnetic interference characteristics.- The equipment shall conform to class 1C equipments specified in MIL-STD-461.

3.8 The equipment shall be in accordance with the supplementary requirements specified in MIL-E-16400.

3.9 Drawings and manuals.- When required by contract or order, instruction manuals shall be type 1, in accordance with MIL-M-15071, and installation drawings shall be in accordance with MIL-D-23140.

3.10 Safety.- Safety shall be in accordance with Requirement 1 of MIL-STD-454.

3.11 Workmanship.- Workmanship shall be in accordance with Requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality assurance system.- The contractor shall provide and maintain an effective inspection and quality control system acceptable to the Government covering the supplies under the contract. The quality control system shall be in accordance with MIL-Q-9858. A current written description of the system shall be submitted to the command or agency concerned via the cognizant Government inspector, for system approval, 60 days prior to first article inspection. If first article inspection is not required, the written description of the quality control system shall be submitted for approval 60 days prior to production inspection. An advance copy shall be forwarded to the command or agency concerned. The written description shall include flow charts showing the complete flow of material from inspection on receipt, through all manufacturing processes, to final shipment. The various processes and inspection positions shall be clearly identified on the flow charts and cross referenced to the detailed inspection procedures and criteria of the written description of the quality control system. Any changes to the approved quality control system which might affect the degree of assurance required by this specification or other applicable documents shall be submitted to the command or agency concerned for approval prior to use.

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4.1.2 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, and (b) Government product inspection to measure quality of product to be offered for acceptance. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Tests, test equipment, and test procedures.- The contractor shall submit a list of proposed tests and written test procedures including test equipment and tolerance limits to be used in testing first article and production equipment to the command or agency concerned via the Government inspector, for review/ approval, 60 days prior to beginning first article inspection. (An advance copy shall be forwarded to the command or agency concerned.) Each test shall be identified in accordance with 4.2 as to classification. The contractor may, at his option, submit the production portion 60 days prior to beginning production.

4.2 Classification of inspection.- The method of examination and testing of the equipment fall within the following classifications:

(a) First article inspection
(b) Quality conformance inspection
    (1) Production inspection
    (2) Production control inspection
    (3) Environmental inspection

4.3 First article inspection.- First article inspection shall consist of all examination and testing necessary to determine compliance with this specification. First article inspection shall include the inspections specified in table I.

TABLE I. First article inspection

Inspection	Reference paragraph
Production	4.4.1
Production	4.4.2
Environmental	4.4.3

4.3.1 First article inspection report.- Six copies of the first article inspection report shall be submitted within 30 days of completion of the first article inspection to the command or agency concerned, via the Government inspector, for review and approval. An advance copy shall be forwarded to the command or agency concerned. The first article inspection report shall indicate specification compliance limits as well as actual performance test results. The contractor shall certify that, on the basis of the report, the equipment does or does not comply with specification requirements.

4.4 Quality conformance inspection.-

4.4.1 Production inspection.- Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the tests approved in accordance with 4.1.3 and shall include the following:

Inspection	Reference paragraph
Surface examination	4.5.1
Operating test	4.5.2

4.4.2 Production control inspection.- Production control inspection, including sampling, shall conform to table II and to the procedures for special inspection levels of MIL-STD-105. The inspection level shall be

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S-4 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. The equipments shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

TABLE II. Production control inspection

Inspection	Reference paragraph	AQL (percent)
Weights and dimensions	3.6.2	15
supply line voltage and frequency	MIL-E-16400, 4.5.4	6.5
Heat test	MIL-E-16400, 4.5.6	6.5
Power	3.5.1, 3.5.2.1	6.5
Electromagnetic interference	3.7 and MIL-E-16400, 4.5.11	15
Controls and control circuits	MIL-E-16400, 4.5.12	6.5

4.4.2.1 Rejected lots.- If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental inspection.- Environmental inspection shall be conducted to prove the durability of the equipment to withstand the effects of extremes of environment, service conditions, and hard usage. Environmental inspection shall be performed on equipment or equipments that have passed production inspection as specified in 4.4.1 and shall include the tests listed in table III.

TABLE III. Environmental inspection

Inspection	Reference paragraph
Temperature	3.4.3 and MIL-E-16400, 4.5.8
Humidity	3.4.3 and MIL-E-16400, 4.5.9
Vibration	4.5.3
Shock	4.5.4

4.4.3.1 Environmental inspection samples.- Samples for the environmental inspection shall be selected without regard to their quality in accordance with the following:

(a) One sample from the first 25 production units fabricated.
(b) One sample from each successive 100 units produced or fraction thereof.

4.4.4 Nonconformance and retest.- If a sample unit fails a test specified in 4.4.2 or 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken to correct units of product which were manufactured with the same conditions, materials, and processes: the equipment shall then be retested.

4.4.5 Corrective action.- If corrective action results in a change of part(s), circuit layout, or the mechanical arrangement, or if the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred immediately to the contracting officer. It shall then be determined by the command or agency concerned whether correction should be made to the equipments already delivered under the contract.

4.4.6 Reinspection of conforming production control and environmental test units.- Unless otherwise specified, sample units which have been subjected to and passed production control and environmental inspections may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all visible damage.

4.5 Inspection procedures.-

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4.5.1 Surface examination.- Equipment shall be examined for workmanship, assembly and fit, marking, materials, parts, and finish for conformance to the requirements of this specification.

4.5.2 Operating test.- The equipment shall be energized and subjected to an 8-hour operating test under full load in normal room temperature to ensure qualitatively the proper functioning of the equipment, including all operating controls. This test shall include, but not be limited to, steps to verify compliance with the requirements of 3.5.2, 3.5.3, 3.5.4, 3.5.5, 3.5.7, 3.5.8, and 3.6.6.

4.5.3 Vibration.- Equipment shall be subjected to the vibration test of MIL-STD-810, Method 514, Procedure XI.

4.5.4 Shock.- Equipment shall be subjected to the shock test of MIL-STD-810, Method 516, Procedure II.

4.6 Inspection of preparation for delivery.-

4.6.1 Level A packages.- Inspection of level A packaging shall be made to determine compliance with MIL-P-116.

4.6.2 First article.- First article inspection, including the rough handling test, is required except where level C packaging and packing is specified. First article inspection will not be required when such a package and pack has previously been inspected and accepted for the same method for a similar item, and satisfactory evidence can be furnished to the Government that items have been prepared identically with the previously approved package and pack.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Packaging.- Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Units.- Each unit shall be packaged in accordance with submethod IIb of MIL-P-116. The package, including the interior parts of the unit, shall meet the physical protection requirements of MIL-P-116. Containers interior and exterior to the barrier shall conform to PPP-B-636, class—domestic. Desiccant shall conform to MIL-D-3464, type II or III.

5.1.1.2 Technical literature.- Technical literature shall be packaged in accordance with submethod IC-1 of MIL-P-116 using transparent barrier material conforming to MIL-F-22191, type II. The technical literature package shall not be placed within the barrier material used to package any other item.

5.1.1.3 Equipment repair parts.- Equipment repair parts shall be individually packaged in transparent packages and be provided with repair parts lists in accordance with MIL-E-17555.

5.1.1.4 Consolidated package.- Items comprising a complete unit and packaged in accordance with the method of preservation specified shall be consolidated in containers conforming to PPP-B-636, V3. The technical manual package shall be placed on the top of the unit prior to closure. Containers shall be closed and sealed with tape minimum 2 inches wide conforming to PPP-T-76.

5.1.2 Level C.- Packaging of individual units shall be sufficient to afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity for immediate use.

5.2 Packing.- Packing shall be level A, B or C, as specified (see 6.2).

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5.2.1 Level A.- Units packaged as specified in 5.1 shall be packed in containers conforming to MIL-B-43291 or PPP-B-640, class 2. Containers shall be reinforced with tape conforming to PPP-T-97, type IV. The gross weight of packs shall not exceed 100 pounds.

5.2.2 Level B.- Units packaged as specified in 5.1 shall be packed in containers conforming to PPP-B-636, Class—weather-resistant, special requirements, or PPP-B-001163. Closure and reinforcement shall be in accordance with appendix of the applicable box specification, except that only nonmetallic tape banding shall be used for reinforcement.

5.2.3 Level C.- Units packaged as specified in 5.1 shall be packed in containers conforming to PPP-B-636 and in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable. Individual sets packaged as specified in 5.1.1 require no further packing.

6. NOTES

6.1 Intended use.- The equipment covered by this specification is intended for military service at shore stations where it is expected to withstand continuous use for long periods, under military service conditions, without benefit of overhaul. The equipment is in each case a vital instrument; failure at a critical moment invariably results in serious reduction in the communications efficiency. Emergency repairs can seldom be made with sufficient celerity to avoid having equipment out of service during a critical period.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Levels of preservation, packaging, packing, and marking (see section 5).

6.3 First article.- Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - EC
(Project 5815-N053)

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